

Operational Diagnostics Report

Kenyan Fuel Depot Pipeline Operations — Year 2026 Throughput Investigation | Prepared for the Operations Director

Context: What We Were Asked to Look Into

We were handed a year of shift-level operating data across our four depots — Mombasa, Kisumu, Eldoret, and Nairobi — covering pump throughput (barrels moved per shift), maintenance status, weather, and grid voltage. The question: is there a real operational problem hiding in this data, and if so, where and why?

The short answer is yes — and it is unusually easy to fix, because it is not spread across the business. It comes from one pump.

Insight: The Root Cause

Nairobi depot is running well below the other three depots, but only because of one of its four pumps — NBI-P03. Nairobi's other pumps perform in line with the rest of the fleet. NBI-P03 itself performs normally when it has been maintained (1,052 barrels/shift on its clean shifts, above the company average). The problem is that it almost never has been: 529 of its 566 logged shifts — 93% — are flagged as running with missed maintenance, and those shifts average 625 barrels, a 40% drop.

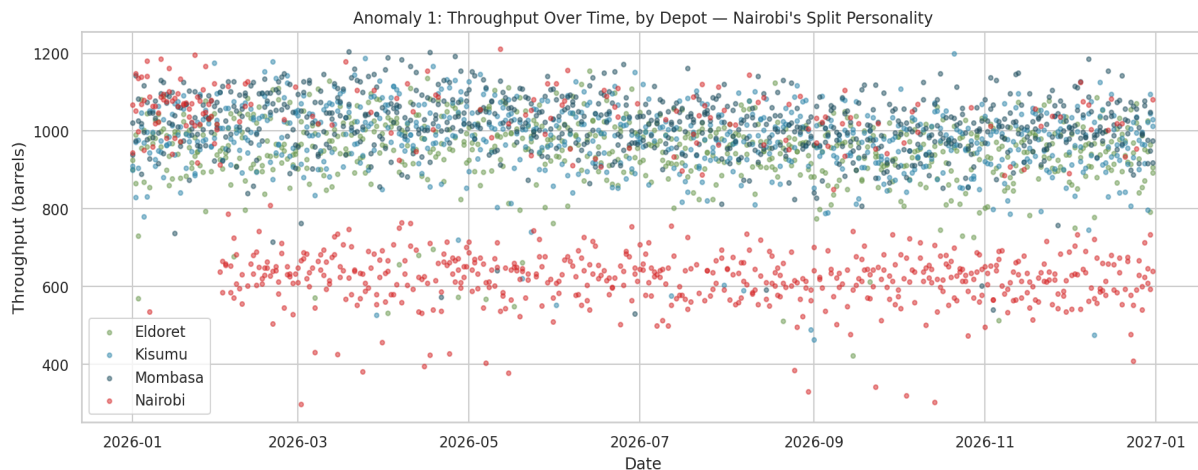


Figure 1. Nairobi (red) is the only depot that splits into two bands. The lower band begins in February and holds steady through year-end.

We quantified this with a Pareto analysis: attributing every shift's throughput shortfall (versus the fleet's normal-operation baseline) back to its pump. **NBI-P03 alone accounts for 92.8% of all estimated throughput loss across all four depots and thirteen pumps combined** — roughly 194,000 barrels over the year.

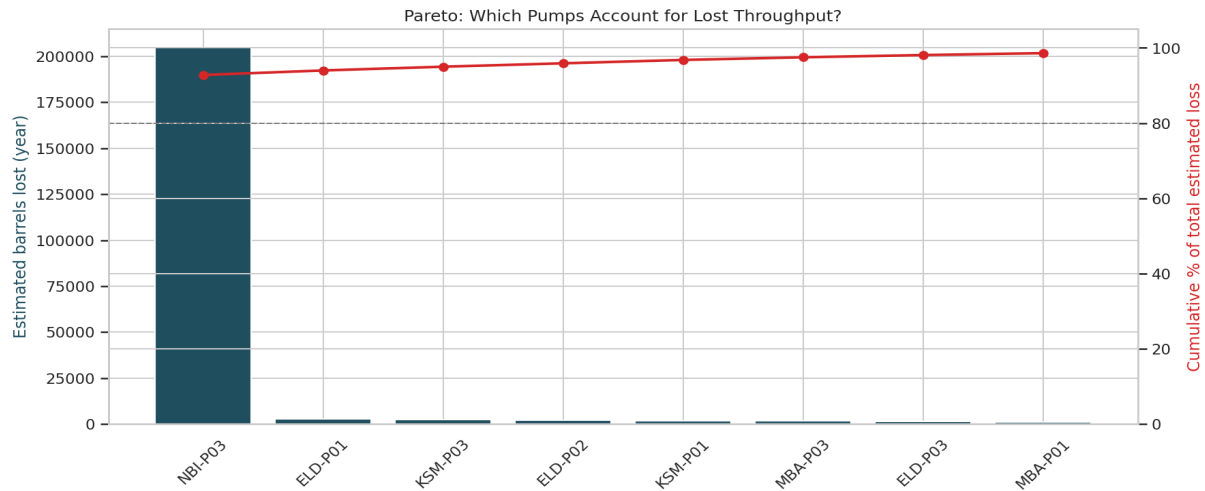


Figure 2. Estimated lost throughput by pump. One bar dominates the entire year's loss.

Before pinning this on maintenance, we checked the two obvious alternative explanations. **Weather:** correlation between temperature and throughput is +0.25 — warmer days trend slightly higher output, the opposite of what a “heat is hurting the pumps” theory would predict. **Grid voltage:** correlation is +0.04, essentially zero. Both are ruled out. **Maintenance status** correlates at -0.85 — the strongest relationship in the entire dataset, and the only one that actually explains what we’re seeing.

Action: What We Recommend

1. Ground NBI-P03 for its overdue service immediately

It is the single highest-leverage fix available anywhere in the operation right now — closing this one gap recovers an estimated ~194,000 barrels/year, without touching any other depot or pump.

2. Find out why this pump alone is skipping maintenance

Every other pump in the fleet is maintained on schedule; this one, sitting at 566 shifts versus a fleet average of ~225, is likely being run continuously to cover a gap (short-staffing, another pump down, demand surge) and maintenance keeps getting deferred as a result. Fix the scheduling pressure, not just this month’s backlog.

3. Add a simple maintenance-overdue alert, not a bigger dashboard

This didn’t require complex analysis to catch — a shift counter or days-since-last-service threshold on each pump would have flagged NBI-P03 months ago. Let the data do that going forward instead of waiting for a year-end review.

Bottom line: this is not a fleet-wide reliability problem requiring a broad program. It’s one pump, one root cause, and one fix — with the data to prove it and a clear before/after metric (NBI-P03’s own “no-incident” shifts) to confirm it worked once maintenance resumes.